

HADOOP INSTALLATION

Windows 11/10 → WSL2 → Ubuntu → Java 11 → Hadoop 3.4.2

1. Prerequisites

Processor	Intel Core i5 or equivalent
RAM	Minimum 8 GB
Free storage	Minimum 100 GB free space
Virtualization	Must be Enabled in Task Manager → Performance → Virtualization

IMPORTANT: Before installing WSL, open Task Manager → Performance and verify that “Virtualization” is Enabled.

2. Install WSL and Ubuntu

Open PowerShell as Administrator and run the following command:

```
wsl.exe --list --online
```

Check whether Ubuntu-24.04 is available in the list.

```
wsl --install Ubuntu-24.04
```

Alternative: If Ubuntu-24.04 is not available in the list, use:

```
wsl --install Ubuntu
```

if it prompts to restart then restart the system and open ubuntu in your system

3. Complete the Ubuntu First-Time Setup

A successful installation will display messages similar to:

```
Downloading: Ubuntu 24.04 LTS
Installing: Ubuntu 24.04 LTS
Distribution successfully installed.
It can be launched via 'wsl.exe -d Ubuntu-24.04'
Launching Ubuntu-24.04...
Provisioning the new WSL instance Ubuntu-24.04
```

Create the default Unix user account:

```
Create a default Unix user account: ubuntu
New password: 1234
Retype new password: 1234
passwd: password updated successfully
```

The password will not be displayed while typing in Ubuntu.

4. Update Ubuntu Packages

```
sudo apt update  
sudo apt upgrade -y
```

5. Install Java JDK 11

Install OpenJDK 11:

```
sudo apt install openjdk-11-jdk -y
```

Verify the Java installation:

```
java -version
```

Expected result: Java 11 should be shown in the version information.

6. Download Hadoop 3.4.2

Go to the home directory first:

```
cd ~
```

Download Hadoop 3.4.2:

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.4.2/hadoop-3.4.2.tar.gz
```

7. Extract Hadoop

```
tar -xzf hadoop-3.4.2.tar.gz
```

8. Move Hadoop to /usr/local

```
sudo mv hadoop-3.4.2 /usr/local/hadoop
```

Verify that Hadoop files are present:

```
ls /usr/local/hadoop
```

9. Configure Environment Variables

Open the Bash configuration file:

```
nano ~/.bashrc
```

Go to the bottom of the file and add the following:

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export HADOOP_HDFS_HOME=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

Save and exit nano:

```
Ctrl + O    → Save
Press Enter → Confirm
Ctrl + X    → Exit
```

10. Apply the Configuration

```
source ~/.bashrc
```

Ctrl + X is only used to exit the nano editor.

11. Verify Hadoop Installation

```
hadoop version
```

Expected output:

```
Hadoop 3.4.2
```